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Barriers to opioid use disorder treatment among people who use drugs in the rural United States: A qualitative, multi-site study

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CRediT Author Statement

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Declarations

Ethics approval and consent to participate

All research was conducted in accordance with both the Declarations of Helsinki and Istanbul, all research was approved by the appropriate ethics and/or institutional review committee(s), and written consent was given in writing by all subjects.

Consent for publication

Not applicable

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Abstract

Background: In 2020, 2.8 million people required substance use disorder (SUD) treatment in nonmetropolitan or ‘rural’ areas in the U.S. Among this population, only 10% received SUD treatment from a specialty facility, and 1 in 500 received medication for opioid use disorder (MOUD). We explored the context surrounding barriers to SUD treatment in the rural United States.

Methods: We conducted semi-structured, in-depth interviews from 2018–2019 to assess barriers to SUD treatment among people who use drugs (PWUD) across seven rural U.S. study sites. Using the social-ecological model (SEM), we examined individual, interpersonal, organizational, community, and policy factors contributing to perceived barriers to SUD treatment. We employed deductive and inductive coding and analytical approaches to identify themes.

Results: Among 304 participants (55% male, mean age 36 years), we identified barriers to SUD treatment in rural areas across SEM levels. At the individual/interpersonal level, relevant themes included: fear of withdrawal, the need to “get things in order” before entering treatment, close-knit communities and limited confidentiality, networks and settings that perpetuated drug use, and stigma. Organizational-level barriers included: strict facility rules, treatment programs managed like corrections facilities, lack of gender-specific treatment programs, and concerns about jeopardizing employment. Community-level barriers included: limited availability of treatment in local rural communities, long distances and limited transportation, waitlists, and a lack of information about treatment options. Policy-level themes included insurance challenges and system-imposed barriers such as arrest and incarceration.

Conclusion: Our findings highlight multi-level barriers to SUD treatment in rural U.S. communities. Salient barriers included the need to travel long distances to treatment, challenges to confidentiality due to small, close-knit communities where people are highly familiar with one another, and high-threshold treatment program practices. Our findings point to the need to facilitate the elimination of treatment barriers at each level of the SEM in rural America.

Introduction

In 2020, 2.8 million people met the criteria for a substance use disorder (SUD) in nonmetropolitan or ‘rural’ areas in the United States (U.S.).¹ Yet, among people with a SUD, only 10% received substance use treatment from an inpatient or outpatient program, a hospital, or mental health program, and 1 in 500 received medication for opioid use disorder (MOUD).¹ People living with opioid use disorder (OUD) who do not access treatment face higher risks of adverse outcomes including fatal overdoses, hepatitis C virus and HIV infections, and poorer quality of life.^{2–5} These risks are more difficult to prevent in rural

areas due to heightened barriers such as highly stigmatized perceptions of opioid use, long distances, inadequate transportation, and lower provider-to-patient ratios compared to urban settings.^{6,7} It is imperative to understand and address barriers to OUD treatment among rural people who use drugs (PWUD) to reduce opioid-related morbidity and mortality.

Existing high-threshold treatment program practices and government policies harm people with OUD by worsening treatment and illness burdens, compounding life demands, and straining resources, resulting in low rates of SUD treatment uptake.⁸ A prior study identified barriers to SUD treatment at various levels of the social-ecological model (SEM) in an urban setting.⁹ Individual-level barriers include life circumstances, negative experiences with MOUD, and treatment readiness.^{9,10} Organizational and community-level barriers include stigma from healthcare providers and employers, long treatment waitlists, and strict program rules.^{9–13} Societal-level barriers include social norms (e.g., morality), stigma tied to substance use, system-level policies, and zoning laws against treatment centers.^{14–16} While barriers to treatment are well-documented among the general population of PWUD, little is known about the unique barriers rural PWUD experience in accessing and remaining in OUD treatment.¹⁷

As part of the larger multi-site Rural Opioid Initiative (ROI), we used a multidimensional approach to better understand barriers to OUD treatment among PWUD in the rural U.S. across the following levels of the SEM framework: individual/interpersonal, organizational, community, and policy.

Methods

Study Design

The U.S. Department of Health and Human Services and the Appalachian Regional Commission funded the ROI to better understand and respond to the effects of the opioid crisis in the rural U.S., specifically eight studies spanning 10 states and 65 counties across the country.¹⁸ The overarching goal of the ROI was to help rural communities develop comprehensive approaches to prevent and treat substance use-related outcomes.¹⁹ We aimed to recruit and enroll PWUD from a diverse mix of rural communities. ROI studies conducted research through two phases. The first phase involved, epidemiologic and policy scans and collection of harmonized qualitative and quantitative data to facilitate comparisons and aggregation of data across sites. The second phase focused on data-informed interventions. The goal of the current paper is to describe qualitative findings from Phase 1, with a focus on barriers to OUD treatment.

Rurality was defined using the US Health Resources and Services Administration “Am I Rural” website (<https://www.ruralhealthinfo.org/am-i-rural>), which relies on definitions of rurality from a number of US federal programs. This “ensembled” approach to defining rurality relies on a number of measures: population count, population density, housing unit counts, and commutes from places of residence to nearby urban areas. ROI-funded agencies used this website to inform site selection.

The rural sites in Northern New England included contiguous communities that straddle the Connecticut River Valley in Massachusetts, Vermont, and New Hampshire. Rural study sites in Kentucky, North Carolina, West Virginia, and Ohio, were located in Appalachia. The rural sites in midwestern US counties were located throughout Wisconsin and in Southern Illinois. The Oregon sites in the Pacific Northwest included both inland and coastal communities in expansive, sparsely populated counties. The populations in these areas are predominantly White, with an estimated 12–38% of the population living below the poverty line, compared to the U.S. average of 10.5%. These rural communities have experienced elevated vulnerabilities related to opioid overdose, as well as HCV and HIV transmission, due to high rates of injection drug use and subpar healthcare systems.^{20–22}

Overall participant eligibility is described in detail elsewhere,¹⁸ but all participants were at least 18 years of age and reported past 30-day injection drug use or nonmedical opioid use. PWUD were recruited between 2018–2019 to participate in 60-to-90-minute interviews. Following informed consent and interview completion, each study paid participants, with compensation ranging from \$25–\$50.

A semi-structured interview guide was developed and harmonized by investigators across ROI sites. The interviews covered a range of topics, including substance use history, risk practices, and overdose experiences; sexual partners and risk practices; criminal legal history; OUD treatment history, access, and experiences; and access to and attitudes toward mental health and health care services (Appendix 1). For this qualitative study, we focused on data from Kentucky (KY), Illinois (IL), Massachusetts (MA), New Hampshire (NH), North Carolina (NC), Ohio (OH), Oregon (OR), Wisconsin (WI), and Vermont (VT).

Qualitative Analysis

Qualitative research team members with experience in anthropology, epidemiology, public health, addiction medicine, harm reduction, and social services conducted in-person interviews at each site. Interviews were audio recorded and transcribed verbatim. The core analytical team included evaluators with expertise in epidemiology (TS, AE), anthropology (RF), substance use research (TS, WZ, AE, GL, JS, JB, SW, SM, ER, PD), harm reduction (PD), public health (WZ), and addiction medicine (PF). The Qualitative Core of the University of Washington Data Coordinating Center (UW-DCC) redacted identifying information from the transcripts, which were identified only by a unique identification number. After verification of transcript accuracy, audio recordings were erased. UW-DCC utilized Dedoose content analysis software for data management and analyses.²³ Preliminary deductive coding was conducted to label emergent themes per the questions in the interview guide. Data were then inductively analyzed for emergent themes within each code, using a data-driven thematic coding scheme to generate refined codes. This process continued iteratively until thematic saturation was achieved, and the organization of the conceptual framework was stabilized. Evaluators independently applied the refined codes to identical data and met regularly to ensure concordance in code applications. Overall and site-specific code applications were summed to identify the most salient codes based on frequencies and percentages. Subsequent code report data were reviewed by two evaluators and memos were applied to highlight salient themes. Discrepant themes were discussed and resolved. Regular

analytic team meetings were held to discuss quality assurance and inter-rater reliability issues.

Quantitative Analysis

We developed a harmonized core survey instrument collaboratively and included items from multiple domains: demographics, drug use practices, infectious disease history, sexual behaviors and risk factors, overdose, and access to harm reduction and addiction treatment services. Study sites used an audio computer-assisted self-interview (ACASI) or computer-assisted provider interview (CAPI). Measures were selected based on prior research and existing instruments. The survey included items focused on barriers to OUD treatment overall and by MOUD, detoxification, and residential treatment. We calculated frequencies and percentages for overall and site-specific barriers.

SEM Framework

Due to the complex network of factors that impact access to OUD treatment in rural settings, we contextualized our findings using the SEM. The original SEM included five environmental systems that extend from an individual's interactions with their environment from a microsystem (family/peers) to macrosystem (cultural attitudes and ideologies).²⁴ The SEM has been adapted and applied to contextualize factors that prevent access to and engagement with treatment in an urban northeastern city⁹ and to demonstrate the multi-layer complexity of the opioid crisis.²⁵ We condensed the original model to four levels: 1) individual/interpersonal, 2) organizational, 3) community, and 4) policy (Figure 1). While many themes are specific to one level of the SEM, several are relevant across multiple levels, indicated by connecting arrows across themes and levels.

The study protocols for each ROI site were reviewed and approved by each site-specific institutional review board.

Results

The 304 participants we interviewed from New England, Appalachia, the Midwest, and the Pacific Northwest were 55% male, 70% white, and had a mean age of 36 years. Eight in ten participants (80%) reported injecting drugs during the 30-days prior to their interview, and 18% completed less than a high school education (Table 1). We identified perceived barriers to OUD treatment in rural areas across all levels of the SEM.

While we did not note salient site-specific OUD treatment barriers in our deductive and inductive analyses, in counting code applications across our large qualitative dataset, we noted some differences in code applications by type of treatment barriers (Supplemental Table [ST] 1). For inpatient treatment, distance appeared to be a bigger concern for IL and NE, and cost appeared to be more of a barrier in KY and NE. For residential treatment, distance appeared to be a significant barrier for NC. Cost appeared to be more of a barrier for NC and OH. However, site-specific denominators for our qualitative interviews may be too different to meaningfully compare frequencies across sites. It was unclear, for instance, whether a 20% prevalence of a barrier in New England (n=22) meant the same as a 20% prevalence in NC (n=65). In addition, our survey data frequencies by site indicated that

there was minimal variation in OUD treatment access by treatment type, with the following exceptions: WI and KY appeared to have worse buprenorphine access, while NC and OR had better access. WI appeared to have worse access to every kind of OUD treatment. Specific barrier-oriented questions were only included in the survey for MOUD treatment. No meaningful differences were noted between sites (ST 2).

Individual and Interpersonal Level Barriers

Rural PWUD noted several person-level themes for not engaging in OUD treatment. These included: motivation for treatment, fear of withdrawal, the need to “get things in order” before entering treatment, interpersonal networks and settings that perpetuated use, and stigma. Motivation or readiness was noted as a prominent barrier to MOUD treatment (n=52 of 304 participants; 17%), detoxification (n=57/304; 19%), and inpatient treatment (n=112/304; 37%), with consistency across sites (ST 1).

Motivation for Treatment

At the most basic level, participants who felt they were not ready to enter treatment noted two primary reasons: The belief that they needed treatment but were not ready to initiate or that abstinence wasn’t achievable.

I’m not ready to go. I’m just not ready. I need to go and get the shit out of my system and ... I’m an addict. I don’t think I’m ever going to be able to quit. (Male, 33, NC)

Need to ‘Get things in Order’ Before Entering Treatment

Several participants noted their life circumstances and responsibilities as a barrier to treatment. Participants reported efforts to get their jobs, houses, finances, health issues, and family affairs ‘in order’ before being ready or able to enter treatment.

I would [get another MOUD prescription.] Now that I’m starting this new job... after my nine-day probationary period at this job, I will be given the option to be completely covered on anything. I’ll still have to pay some doctors and stuff, but it’ll be way less, so I’m thinking about going back and finding a Suboxone program to partake in once I get the medical insurance through my new job...to completely stop using opiates. (Male, 24, NC)

So, I guess, I’m just doing the best I can, what I got right now. I’ve been down this road so many times, and, I mean, I know how it ends if I keep going the way it’s going, so I’m just trying to maintain my job and my apartment and my bills with also trying to, you know, get into it, a detox. I have something, a basic plan, set up, I just haven’t put it in complete motion yet. (Female, 25, WI)

Fear of Withdrawal Symptoms

Participants commonly cited fear and avoidance of opioid withdrawal as a barrier to OUD treatment based on their lived experience or perceptions rooted in their peers’ experiences.

I know a guy who's on Vivitrol, and he's been 5 years on Vivitrol...I mean I've thought about it, but you've gotta go those 7 days and about your 3rd day is when it really starts hitting you and you come into withdrawals. (Female, 42, OH)

I've thought about [detox/drug treatment] a couple times, but I never went. The thought of being sick, thought of not being able to—detox, coming off the drugs [prevented me from entering treatment]. (Male, 28, KY)

Interpersonal Influences, Settings, and Triggers that Perpetuate Drug Use

With rural areas being home to smaller, tight-knit connections, participants noted the struggles of returning to their everyday settings following treatment, which perpetuates drug use. Participants recognized the importance of changing people, places, and things to avoid triggers that might lead to re-initiation of use. Long-term residential treatment was noted as particularly appealing by participants who needed to move away from potential triggers and interpersonal relationships in their rural communities, where everyone knows one another:

I know, I been there, they said first thing is change people, place and things. I try to do that but that ain't enough, man...I need help...I just can't do outpatient, I get out and see these motherfuckers, I'm gonna get high again. It don't work for me, I need to get my mind right...You know what I'm saying because I need the money to survive, dude...A lot of people are up on this hill. Man, there's people selling, there's people doing and it's all because of lack of everything. You know lack of activities, transportation, jobs, just everything. It's a shithole on earth, man, no joke, dude, if you're an addict because there's just a big circle, man. (Male, 57, IL)

Additionally, participants alluded to substance use-related stigma among clients, family, friends, and general community members, which is exacerbated in their tightknit rural networks. Examples of stigma included feeling ashamed to seek treatment, concerns about being known as a “drug user type,” or being judged by others in Narcotics Anonymous groups.

Yes. I mean there's been times in the past where I thought I needed rehab and I didn't go. There's been times that I was pretty sick and didn't go either. I think both incidents were because of shame. I didn't want to talk about it. It wasn't that I didn't want to quit, it was just I was too embarrassed to talk about it. (Male, 38, NC)

Participants experienced stigmatization from prison employees, community members, and extended family members. One participant noted that “people finding out” and the associated fear of others' awareness stopped them from entering treatment, as did the perception within interpersonal relationships that MOUD replaces “one drug for another.”

Organizational Level Barriers

Several organizational-level barriers arose from the data: strict facility rules, treatment programs managed like corrections facilities, lack of gender-specific treatment programs, and concerns about jeopardizing employment. Strict rules were particularly salient among participants as barrier to detox programs (n=61/304; 20%) and inpatient treatment

(n=45/304; 15%). While there was substantial consistency across sites, strict rules were especially concerning for detox facilities among participants in IL (36%), KY (33%), and OR (29%). And strict rules tied to inpatient treatment programs were particularly concerning in NE (59%) and IL (50%) (ST1).

Strict Facility Rules

Strict facility policies were a common barrier among PWUD in rural areas. Participants noted that some facilities required patients to test positive or negative for specific drugs. Participants who reported using benzodiazepines, for instance, found it challenging to identify a treatment facility that would accept them:

No, I've never been. I take prescription Xanax, and detox won't...no detox or rehab will take you with [benzodiazepine] in your system at all...Like, you can't even go...whenever you go into detox you have to fail your drug test to be able to get into detox, but you can't fail it for benzos...So you have to detox from [benzos] on your own and then come in. (Female, 27, IL)

A reported barrier to accessing residential treatment was a requirement to complete a detoxification program or provide a negative urine drug screen prior to residential treatment. Conversely, respondents reported that some treatment agencies (especially MOUD providers) required a positive urine drug screen at intake:

In the last two years I've been to 17 different funerals because of overdoses, and that's ridiculous, but it's, and a few of those overdoses were people who were having a really, really hard time keeping clean, staying clean. They wanted to...get help...but the clinics and the rehabs require you're dirty to go into them. And so, I know of at least four people that only bought and used that day to get into the treatment facility, and it ended up killing them. (Female, 31, NE)

Programs Managed Like Correctional Facilities

Participants reported that SUD treatment facilities often felt like correctional facilities due to the use of punitive approaches and inflexible rules. Participants expressed concern about losing their freedom and agency within treatment facilities. Programs generated anger among participants, which inadvertently contributed to the need to self-medicate with illicit substances:

...it kinda scared you, just the rules themselves, there's a lot of them. Some of them were more free and the environment was in there to help you, not judge you. And then there was a lot of them that was kinda like a jail. They made you feel like you're a really bad thing, more or less punishment. It makes [clients] more angry, it makes them want to rebel even worse, or it makes them so angry that they self-medicate even worse. I know I did a couple of times. And if the ones who come at me, if they cared and was there to help me, that's when I started getting a conscious. (Female, 34, NC)

Participants noted that some facilities were well known for putting little effort into uplifting patients and felt they would have better results if facilities were run by staff with more

caring, rather than punitive, attitudes. They also believed they would be more successful in treatment if the programs were run by people with lived experiences:

I would want it run by ex-addicts and people that went to school that used to be addicts that have went and got degrees for drug counseling and all that. Somebody that's been there and done that, they'll have more understanding and more focus to help you. (Male, 39, NC)

Lack of Gender Specific Treatment Programs

Participants reported a limited number of beds available for women seeking inpatient or residential treatment, making treatment access more challenging for women.

But if you're a guy and need to go...I remember numerous times my brother just called and he was in that night...it's so much easier being a guy getting into rehab. (Female, 27, IL)

Some participants felt gender-specific facilities were needed, suggesting that a lack of such facilities may be a barrier to accessing meaningful treatment, preventing them from seeking treatment or requiring them to find a gender-specific facility that would allow them to focus on recovery and do "what I'm supposed to do."

I think if there's girls in your same unit, you're not going to pay attention to what you need to do...if they allow you to be with other sex, it's just not going to work...so I chose [Inpatient Treatment Facility] because you're separate and you have, you only can really focus on yourself. (Male, 28, NE)

Concerns About Jeopardizing Employment

PWUD found themselves in a conundrum: they wanted to remain in outpatient treatment but experienced concerns about jeopardizing employment and an inability to take time off work for treatment. Participants often reported work as a priority over treatment due to their income needs, as "you can't make money while sitting in [treatment]." Employed participants could not take a leave of absence for treatment due to fears they would experience substance use stigma and retaliation from their employers and co-workers. Without work-sensitive options for treatment, participants felt the need to try to stop using without the support of treatment to remain employed.

A lot of times I try to get through and do [detox] my own way though, especially if I have a job at the time and I don't want to jeopardize the job...It's kind of a catch 22, because when you jeopardize the job by being on something while trying to taper off whatever, so you lose it. If you have a job and you leave to go to detox you could lose...Or stay there and try to do it yourself without doing it, where you're just dealing with the lifestyle and keeping the job. Being a functional addict. So, either way can be hard. (Male, 39, NC)

Community Level

The most salient community-level themes that emerged across rural sites included: limited availability of treatment options in local communities leading to long travel distances,

waitlists, and a lack of information about local treatment options. Long distances were noted as a barrier to MOUD programs (n=29/304; 10%) and inpatient treatment (n=30/304; 11%). Distance as a barrier to MOUD programs was especially prominent in KY (16%), NE (18%), and NC (15%), while distances to inpatient treatment were most salient in IL (27%) and NE (27%) (ST 1).

Limited Treatment Availability

Due to rurality, participants reported limited availability of treatment within local reach, especially for residential in-patient treatment, but also MOUD and outpatient treatment. This lack of availability created barriers to treatment access, requiring extended travel amidst a landscape with limited transportation options:

I mean, I'd like to go back, but it's three days a week and it's hard for my grandma to take me three days a week and sit there and wait 'cause we live all the way in [community X]. So, she don't wanna drive all the way home and then come back to get me...If you stay there the whole time you get counted. If you're late, they don't count it. (Female, 24, NC)

Waitlists

Participants also reported limited access due to long waitlists for treatment in local rural communities, often waiting several months for treatment slots to open, and limited walk-in treatment:

They can't...Apparently there's huge waiting lists to get in for beds and stuff. And the people that really want to stop are poor, just had a bad dealing of cards in their life are screwed. And it's stupid. I think they should make it way more [accessible]...I feel personally they should make something where it's like you should be able to get seven days' worth of Suboxone at [local pharmacy] or something. (Male, 29, NC)

Similar sentiments were shared by participants ready to seek treatment, whether through detoxification programs, inpatient rehabilitation centers, or MOUD programs, where there were often no open treatment slots and long waitlists:

...I've been on the waiting list. The detoxes are all full. You know, they got 120 beds and there's not a bed available and all the places are full. Always...It's hard to get a bed.

They're like "keep calling". You have to go in, sit there all day and wait, hoping that someone leaves to give up their bed. (Male, 39, NE)

Participants reported that, when at the top of waitlists, their life situations had often changed from weeks or months prior, when they first sought treatment, and that they were subsequently in jail or no longer ready for and seeking treatment. At times, the ability to pay for treatment would also moderate availability, especially for buprenorphine.

If you pay cash, you can get [a] walk-in visit (Suboxone clinic)...if you haven't got any cash you've got a four-month waiting list...I mean four months, man, I'm in a

different place, dude...I don't even remember that waiting list...I'm just strung out on speed again. (Male, 37, KY)

Negative Attitudes and Stigma Towards MOUD

Negative attitudes and stigma related to MOUD treatment options represented additional community-level barriers to treatment in rural communities.

Unless it's end of life, I truly feel like [methadone] should be illegal. Now that there's a new alternative. Or not new, but newer alternatives to Suboxone, Buprenorphine, or whatever...I have gone rounds with our medical director about this one. He preaches about how methadone's this wonderful thing. I've never in my life seen anybody be successful with methadone. Like getting on it, off of it, and staying off. It's so risky with the potential for overdose. Like if you use Benzo's, a lot of times the dosing is too high...I just don't like methadone. I don't think it's ethical quite frankly. (Male, 38, NC)

Some participants also emphasized the addiction and overdose potential of MOUD, while others reflected on their peers' negative experiences with MOUD, highlighting adverse outcomes and withdrawal symptoms.

It's just you're trading one dependency for another, because Suboxone—coming off—is just as bad. Well, not my...I've only taken it very few times, but I hear the withdrawal of methadone is extremely terrible. It's a powerful drug. Sixty milligrams is the minimum to make it as an opium blocker, I believe. Yeah, I think 60 is the minimum as an opiate blocker. One through 50 milligrams is a pain reliever. (Male, 29, OR)

Policy Level

Policy-level themes included cost and insurance challenges and limitations, as well as system-imposed barriers. Costs were noted as a barrier to MOUD (n=28/304; 9%) and inpatient treatment (n=64/304, 21%), with MOUD costs and insurance prominent in most sites, and inpatient treatment costs being a major concern among all participants in IL (100%) and one-third of participants in KY (32%) and NE (36%).

Insurance dictates available treatment

Respondents reported a variety of insurance-related barriers. A common barrier was finding treatment programs that accepted respondents' insurance, including Medicaid or Medicare.

I wasn't able to go to the ones in [city] that had open beds right then because my insurance didn't work there, even though it's in the same state. I guess it was a different local [Medicaid managed care] provider. (Male, 29, OR)

Respondents reported difficulties with losing public insurance once they had attained a stable income, recollecting experiences when they “went back to work and lost their insurance”, or experienced inconsistent coverage through private insurance.

They said I made too much money. If you make over \$11,000, they just shut it off on me, I was going to the methadone clinic and I couldn't pay because I couldn't afford it because they shut my insurance off... (Female, 28, WI)

Other insurance barriers included a lack of coverage for residential treatment or limits on the number of days of detoxification services, with participants noting that "insurance would only cover me for three days", which she found "pointless". And, even when programs could cover longer stays, insurance did not always keep pace:

Yeah, they have, like, 90- to 60-day, like, 3- to 6-month programs or whatever. And I just thought that's what I needed but my insurance is accepted there but only for outpatient, not inpatient, and I obviously feel or it's obvious that I need like intensive, like, inpatient. (Female, 23, WI)

Many respondents could not access insurance and were therefore unable to afford treatment.

When I originally tried getting into the methadone clinic, it was really tough because I didn't have insurance. And if you didn't have insurance, you had to pay something like \$25 a day. Um, and they didn't even want to talk to you if you didn't have the money. (Male, 24, NE)

Even when participants had insurance, and opportunities to move forward with MOUD treatment options, confusion at pharmacies with insurance policies presented additional barriers, as noted by the following participant who tried to get a suboxone prescription filled:

At first it was a bit of a challenge. I have two insurances, I've got Medicare and Medicaid, for the longest time. This is the pharmacy's fault. The pharmacy was constantly getting them confused. Every time I go in to get a prescription filled, I had to tell them to make sure they ran it through my secondary because if they didn't then they'd end up having to contact me and have me tell them again. (Male, 35, KY)

System-imposed barriers

Respondents reported that arrest and incarceration in local jails could interfere with treatment access and continuity, noting that they had been incarcerated while on a waitlist for MOUD treatment, losing their spot when it eventually opened. They also identified challenges to communications between jails and community-based treatment providers, which were often quite distant from one another in rural counties, hindering treatment continuity post-release:

Jail is mostly the number one thing that gets in the way, in and out of jail. If it wouldn't have been for jail, I wouldn't have to re-engage seven times. I wouldn't be kicked out of that place if it weren't for me being thrown in jail. I am supposed to be on Suboxone Tuesday, and if I go to jail, I'll miss that. I can't miss that; I've been waiting on that for four weeks. (Male, 63, OR)

Beyond incarceration, an additional systems-level barrier included finding pharmacies in rural communities that were ready or willing to dispense buprenorphine to all patients with prescriptions.

I guess whenever it comes to Suboxone patients, [pharmacies] will only fill so many prescriptions. I don't know if that's because of something that they're legally obligated to do like a physician is. A physician can't see more than X amount of patients at a time to prescribe them Suboxone. There's a cap on it. I don't know if pharmacies have the same restrictions or not. But a lot of pharmacies won't take new prescriptions from people with Suboxone who haven't gotten it filled there before. (Male, 35, KY)

Policies and regulations related to buprenorphine prescription, including historical requirements for clinicians to receive a waiver to prescribe and caps on the number of patients a provider could prescribe for, also presented barriers to treatment.

The whole Suboxone process is like ridiculous in my opinion. How only certain doctors can prescribe it and only X amount of patients. It's just stupid. If you're trying to stop the opiate epidemic, which is out of control right now, there's so many regulations and stuff. They're making it so hard for the people that want to stop. And it's like I was lucky and fortunate enough to have a full-time job and insurance but the people that don't, they're just fucked basically. (Male, 29, NC)

Discussion

We identified barriers to OUD treatment across all levels within the SEM. At the individual/interpersonal level, unique rural barriers included close-knit communities and tightknit rural networks, where everyone knows one another, making it difficult to evade local substance use recurrence triggers, and stigma, networks and settings that perpetuated opioid use. Organizational-level barriers included: strict facility rules, treatment programs managed like corrections facilities, lack of gender-specific treatment programs, and concerns about jeopardizing employment, in rural communities with high poverty and unemployment rates. Community-level barriers included: limited availability of treatment in local rural communities; the vast distances that one needed to travel in terms of geography, time, and treatment knowledge; limited transportation; and waitlists. Policy-level themes included insurance challenges and system-imposed barriers such as arrest and incarceration.

At the individual-level, a key barrier was the belief that one needed to “get things in order” before entering treatment from a personal, familial, addiction, and work perspective, consistent with findings from urban settings.⁹ Indeed, many current treatment programs are set up to enroll people who are “ready to enter” treatment. Even when participants are ready to enter treatment, waitlists and waiting times often hinder opportunities to initiate treatment.²⁶ These barriers run counter to typical harm reduction-oriented treatment programs that aim to “meet people where they are”.²⁷ “Waiting to get things in order,” a highly subjective and potentially enormous task, places PWUD at high risk for overdose, especially given the rapidly changing risk environment tied to the entry of fentanyl in the drug supply,²⁸ with fentanyl becoming increasingly prevalent in both opioid²⁹ and stimulant drug supplies.³⁰ Korthuis et al. found that among our rural study participants, the co-use of methamphetamine and opioids was associated with higher risks of nonfatal overdose compared to participants who reported using opioids or methamphetamine alone and participants who reported co-use had less success accessing SUD treatment.³¹ Given

the lethality of co-use of methamphetamine and fentanyl, delaying treatment may be fatal. This emphasizes the importance of getting PWUD connected to OUD treatment immediately upon request. However, previous research among stimulant users in rural areas found that less than 30% of them perceived a need for treatment.³²

At the interpersonal level, the tight-knit nature of relationships was magnified in rural communities, which presented barriers to treatment. Factors that increase the recurrence of substance use include sex, age, polydrug use, criminal legal system involvement, affective factors, and comorbid psychological disorders.³³ Rural, tight-knit relationships manifested in difficulties evading potential triggers that led to relapse and judgmental feedback that participants experienced when running into peers or family members. Stigma tied to substance use treatment uptake³⁴ and within treatment programs has been well documented, including negative perspectives of MOUD within 12-step programs,³⁵ perception of MOUD treatment as a “replacement for drugs,”³⁶ and patient and provider judgements and stigmatizing comments and care,³⁷ presenting major challenges. Our findings are consistent with those in other rural areas of the U.S., where “everyone knows everyone” and “everybody knows everybody’s business,”³⁸ which challenges privacy and often hinders one’s willingness to access treatment and care. These barriers contributed to participant interest in getting away from the local scene, with hopes of avoiding triggers and localized stigma in their interwoven communities, and to do so they believed that inpatient treatment was often the answer. However, access to inpatient treatment slots was limited and costs were often prohibitive when such slots were open.

Prominent barriers at the organizational level included strict facility rules, treatment programs run like correctional facilities, and concerns about jeopardizing employment. This fits with the literature regarding the need for low-threshold treatment that provides flexibility.³⁹ Participants expressed a desire for programs that treated them with dignity and respect. The strict rules included requirements tied to treatment regimens, scheduled treatment visits, needing methadone treatment on sight, and urine analysis requirements to substantiate the substances clients had in their systems. Rural participants noted challenges, for instance, accessing methadone treatment if they had benzodiazepines in their system, despite recent research that found that benzodiazepine exposure at intake for methadone treatment did not impact the 12-month treatment retention.⁴⁰ Other participants shared frustration with buprenorphine regulations that have required providers to receive waivers to prescribe and then only to a limited number of patients. A recent study found that, during the COVID-19 pandemic when buprenorphine prescription restrictions were relaxed, there were no negative impacts on opioid-related nor buprenorphine-related overdose risks.⁴¹ Changes to federal regulations in recent years have aimed to increase access to buprenorphine. In 2018, new legislation extended prescribing privileges to other non-physician providers.⁴² In 2021, qualifications among providers to receive a buprenorphine prescribing waiver to treat up to 30 patients were eliminated.⁴³ Finally, in 2023, Congress removed the requirement for health practitioners to apply for the waiver altogether, instead requiring a one-time, eight-hour training, and removing caps on patient numbers.⁴⁴ This is a positive step in diminishing strict OUD treatment rules and legislation, and may offer expanded opportunities for PWUD to access treatment in rural communities. Information campaigns will be crucial to ensure that PWUD learn about these positive policy steps, and newly

opened doors to treatment access. Treatment programs should also change policies requiring positive urine drug screens as pre-entry requirements, diminishing risks for OUD recurrence and overdose among people who have taken steps towards recovery prior to entry into such treatment programs.

The most prominent community-level barriers to substance use treatment among rural study participants included the significant distances that PWUD needed to travel. In urban and rural communities, barriers to treatment access are well documented.⁴⁵ Cross-site results from surveys with participants from our ROI demonstrated the extent of the challenges in rural communities. Within a sample of 3,084 PWUD in rural communities across 10 states, 19% had received any MOUD in the last 30 days,¹⁸ and 40% of the sample tried to access treatment but could not in the past month.³¹ Transportation barriers are accentuated in rural communities, where public transit is limited, at best, and vehicle access is low. Prior studies have pointed to extensive drive-times that PWUD must traverse to get to MOUD programs in rural communities, including travel to adjacent states.⁴⁶ In addition, long waitlists have led to limited access to treatment at times when participants are ready to enter treatment. Faced with waitlists, PWUD continue opioid use to prevent unsupported withdrawal and often lose motivation or experience life events that interfere with treatment entry.^{26,47} Prior studies point to risks of sustained use and lost opportunities to enter treatment while PWUD are on waitlists.²⁶ Interventions and innovations in OUD treatment in rural communities are needed to diminish barriers tied to distance, as well as stigma, which have remained challenging over the past two decades.⁴⁸ The training of primary care providers to provide MOUD⁴⁹ and telemedicine-based treatment has shown promise,⁵⁰ but investment in training and infrastructure to expand such treatment options is greatly needed. Creedon et al. found that telemedicine-based SUD treatment was associated with the receipt of more in-person outpatient services among rural Medicaid beneficiaries, but they indicated that telemedicine-based services are still underutilized.⁵¹ Allowing pharmacists to induce patients with buprenorphine, in collaboration with addiction medicine physicians, has also proven effective, and would be a major benefit to rural areas.⁵²

Salient policy-level SEM themes included insurance challenges and system-imposed barriers. Study participants shared frustrations related to lack of insurance coverage, especially for inpatient treatment, in local rural communities. Participants also decried the variation in the types of insurance that OUD treatment programs accept, limitations in coverage for residential treatment, as well as limited days of coverage for detoxification programs that led them to avoid treatment that would be cut short too soon. While the Affordable Care Act of 2010 led to increases in insurance coverage for mental health and SUD treatment, rural-urban disparities have been noted, with decreased coverage in rural communities.⁵³ Rural study participants noted challenges related to losing public insurance coverage when they returned to work and earned more than the threshold allowed by Medicaid. A nationally representative study found that, even among Medicaid-insured individuals, only 11% had sought treatment during the prior 12 months.⁵⁴

Another reported healthcare system policy barrier was pharmacy limitations on buprenorphine dispensing. Past studies have documented perceived Drug Enforcement Administration (DEA) caps on buprenorphine dispensing among rural pharmacists,^{55,56}

and a recent telephone audit study found that buprenorphine access limitations were particularly common in Southern states.⁵⁷ Pharmacy policy experts have called for DEA policy clarifications and continuing education on this topic with pharmacists.⁵⁸

Incarceration was also noted as a major policy or systems-level barrier to OUD treatment, by cutting treatment short, or diminishing opportunities to finally get off waitlists when slots opened but participants were now incarcerated. Due to legislative mandates, corrections facilities increasingly offer MOUD treatment to incarcerated populations.⁵⁹ Such treatment programs have shown great promise in decreasing OUD and overdose risks post-release from jails; however, continuity of care post-release is hindered by many barriers, including re-initiation of insurance upon release.^{60,61}

Our findings should be considered in light of several limitations. First, while our study includes a diverse mix of rural communities across nine states in New England, Appalachia, the Midwest, and the Pacific Northwest, findings may not be generalizable to other rural communities in the U.S. Second, our data focus on the years just prior to the COVID-19 pandemic, which brought new barriers to treatment, as well as opportunities, with relaxed MOUD prescription policies. Third, the level of probing on barriers specific to different types of OUD treatment may have varied across study sites during qualitative interviews, which could have contributed to differential levels of detail on some treatment barriers. Fourth, our study sample lacked racial/ethnic diversity and future research focused on comparative experiences with OUD treatment barriers among minoritized populations in the rural US is needed. Future research is recommended to assess barriers and facilitators to OUD treatment in rural areas during the COVID-19 era, and a time of rapidly changing drug supplies. Finally, suggested adjustments and new interventions to address treatment barriers will not present a one-size-fits-all opportunity, and tailoring interventions to local rural realities will be necessary.

Conclusion

We collected and analyzed what we believe is the largest and most geographically diverse sample of PWUD living in the rural U.S to date. Our findings highlight barriers to OUD treatment at multiple levels in rural settings. Salient findings unique to rural U.S. communities included the need to travel long distances to treatment, challenges to confidentiality in treatment due to small, close-knit communities where people are highly familiar with one another, a lack of low-threshold harm reduction-oriented treatment practices, and a perception among PWUD that they need to “get their lives in order” before entering treatment. Our findings point to the need to facilitate elimination of treatment barriers at each level of the SEM.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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This publication is based upon data collected and/or methods developed as part of the Rural Opioid Initiative (ROI), a multi-site study with a common protocol that was developed collaboratively by investigators at eight research institutions and at the National Institute of Drug Abuse (NIDA), the Appalachian Regional Commission (ARC), the Centers for Disease Control and Prevention (CDC), and the Substance Abuse and Mental Health Services Administration (SAMHSA). The views and opinions expressed in this manuscript are those of the authors only and do not necessarily represent the views, official policy or position of the U.S. Department of Health and Human Services or any of its affiliated institutions or agencies, or those of the Appalachian Regional Commission. The authors thank the other ROI investigators and their teams, community and state partners, and the participants of the individual ROI studies for their valuable contributions. In addition, the authors especially thank Dr. [Anonymized], our ex-officio steering committee chair, and Dr. [Anonymized], our NIDA scientific research officer, for their thoughtful guidance and leadership over the course of our multi-site study. A full list of participating ROI institutions and other resources can be found at: <http://ruralopioidinitiative.org>.

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Data availability

We welcome collaboration and encourage mentorship and the use of the ROI data stripped of all protected health information (PHI) to enable early investigators to address meaningful questions with support to help ensure their success. Additional information can be obtained at the ROI website: www.ruralopioidinitiative.org or by contacting the ROI DCC at ruralopioidinitiative@uw.edu. Follow the Rural Opioid Initiative on Twitter @ruralopioids.

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Highlights

- We explored barriers to substance use disorder treatment in nine rural US states.
- The social-ecological model (SEM) framed qualitative results from 304 participants.
- We found limited treatment confidentiality abounded in close-knit communities.
- Some rural treatment programs were managed like corrections facilities.
- Findings point to need to eliminate treatment barriers at each level of the SEM.

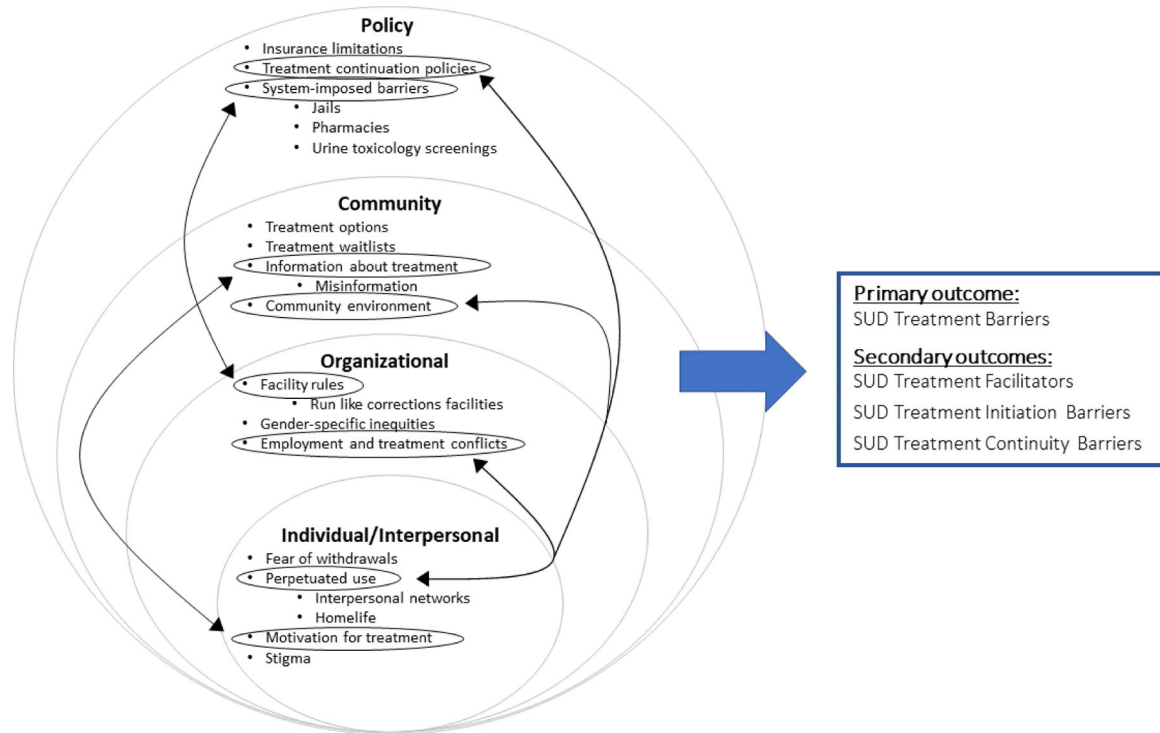


Figure 1. Barriers to OUD treatment in ROI communities in the U.S., 2018–2019. Themes are highlighted at various levels of the social-ecological model (SEM). Bidirectional arrows and ellipses, which encircle some themes, connect salient barriers to OUD treatment that span different levels of the SEM.

Table 1.

Characteristics of rural qualitative interview participants by study sites, 2018–2019 (n=304).

	Characteristics	Total	Illinois	Kentucky	North Carolina	New England	Ohio	Oregon	Wisconsin
Interviewees		304 (100%)	22 (7%)	57 (19%)	65 (21%)	22 (7%)	26 (9%)	52 (17%)	60 (20%)
Male ¹		169 (55%)	14 (64%)	35 (61%)	34 (52%)	10 (45%)	15 (58%)	28 ¹ (54%)	33 (54%)
Mean age		36	37	35	36	33	37	39	35
Race									
	White	213 (70%)	20 (91%)	56 (98%)	15 (23%)	15 (68%)	<i>not asked</i>	49 (94%)	58 (97%)
	Black	2 (1%)	2 (9%)	0 (0%)	0 (0%)	0 (0%)	--	0 (0%)	0 (0%)
	Native American	8 (3%)	0 (0%)	0 (0%)	5 (8%)	1 (5%)	--	1 (2%)	1 (2%)
	Mixed race	5 (2%)	0 (0%)	0 (0%)	2 (3%)	0 (0%)	--	2 (4%)	1 (2%)
	Other	1 (0%)	0 (0%)	1 (2%)	0 (0%)	0 (0%)	--	0 (0%)	0 (0%)
	Not given/Not asked ²	75 (25%)	0 (0%)	0 (0%)	43 ² (66%)	6 ² (27%)	26 ² (100%)	0 (0%)	0 (0%)
Interviewees		304 (100%)	22 (6%)	57 (16%)	65 (19%)	22 (6%)	26 (7%)	52 (15%)	60 (17%)
Substance use									
	Ever IDU	288 (83%)	20 (91%)	57 (100%)	65 (100%)	21 (95%)	17 (65%)	52 (100%)	56 (93%)
	Current IDU	279 (80%)	19 (86%)	57 (100%)	65 (100%)	18 (82%)	15 (58%)	50 (96%)	55 (92%)
	<i>Heroin</i>	195 (56%)	8 (36%)	30 (53%)	51 (78%)	18 (82%)	13 (50%)	41 (79%)	34 (57%)
	<i>Fentanyl</i>	73 (21%)	1 (5%)	17 (30%)	26 (40%)	14 (64%)	3 (12%)	6 (12%)	6 (10%)
	<i>Methamphetamine</i>	210 (60%)	14 (64%)	37 (65%)	61 (94%)	4 (18%)	6 (23%)	47 (90%)	41 (68%)
	<i>Other</i>	193 (55%)	4 (18%)	54 (95%)	61 (94%)	15 (68%)	10 (38%)	21 (40%)	28 (47%)
Interviewees		304 (100%)	22 (7%)	57 (19%)	65 (21%)	22 (7%)	26 (9%)	52 (17%)	60 (20%)
Education									
	< High school	56 (18%)	1 (5%)	21 (37%)	13 (20%)	3 (14%)	7 (27%)	<i>not asked</i>	11 (18%)
	H.S. or GED	97 (32%)	11 (50%)	22 (39%)	21 (32%)	9 (41%)	11 (42%)	--	23 (38%)
	Some college	61 (20%)	7 (32%)	6 (11%)	20 (31%)	3 (14%)	6 (23%)	--	19 (32%)
	Assoc/trade deg	16 (5%)	3 (14%)	8 (14%)	0 (0%)	1 (5%)	0 (0%)	--	4 (7%)
	>= B.A.	12 (4%)	0 (0%)	0 (0%)	8 (12%)	0 (0%)	1 (4%)	--	3 (5%)

	Characteristics	Total	Illinois	Kentucky	North Carolina	New England	Ohio	Oregon	Wisconsin
	Not answered	62 (20%)	0 (0%)	0 (0%)	3 (5%)	6 (27%)	1 (4%)	52 (100%)	0 (0%)

¹One non-male interviewee from Oregon identifies as “Neither”.

²Some NC and NE interviewees did not give their race. Ohio did not ask interviewees their race.